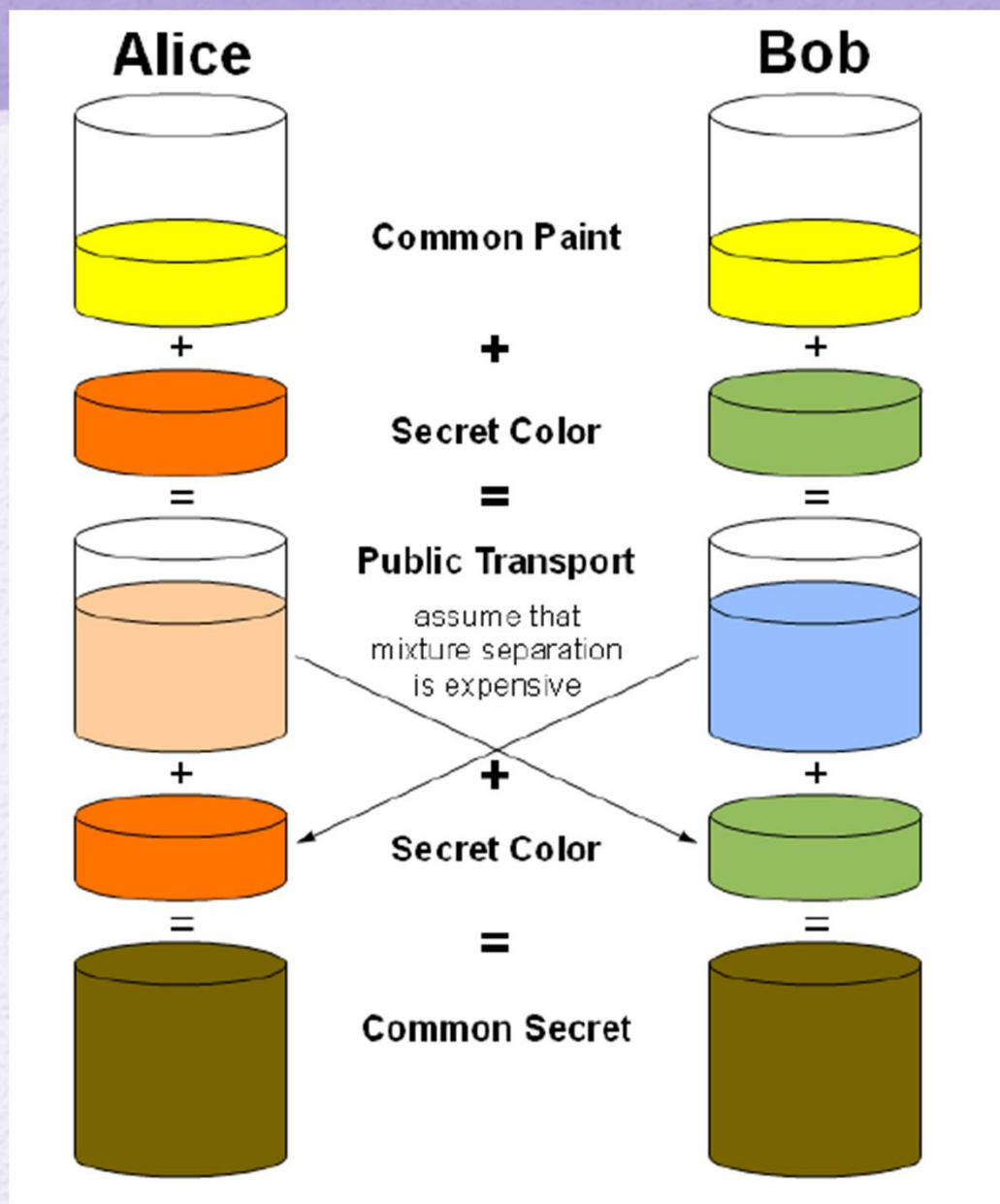


Diffie-Hellman Key Exchange

- First published public-key algorithm
- A number of commercial products employ this key exchange technique
- Purpose is to enable two users to securely exchange a key that can then be used for subsequent symmetric encryption of messages
- The algorithm itself is limited to the exchange of secret values
- Its effectiveness depends on the difficulty of computing discrete logarithms





Alice



Bob

Alice and Bob share a prime q and α , such that $\alpha < q$ and α is a primitive root of q

Alice generates a private key X_A such that $X_A < q$

Alice calculates a public key $Y_A = \alpha^{X_A} \bmod q$

Alice receives Bob's public key Y_B in plaintext

Alice calculates shared secret key $K = (Y_B)^{X_A} \bmod q$



Alice and Bob share a prime q and α , such that $\alpha < q$ and α is a primitive root of q

Bob generates a private key X_B such that $X_B < q$

Bob calculates a public key $Y_B = \alpha^{X_B} \bmod q$

Bob receives Alice's public key Y_A in plaintext

Bob calculates shared secret key $K = (Y_A)^{X_B} \bmod q$



Figure 10.1 Diffie-Hellman Key Exchange

- Diffie-Hellman Example

- **Alice & Bob who wish to swap keys:**
- agree on prime $q=353$ and $a=3$
- **select random secret keys:**
- A chooses $x_A=97$, B chooses $x_B=233$
- **compute respective public keys:**
- $y_A=3^{97} \bmod 353 = 40$ (Alice)
- $y_B=3^{233} \bmod 353 = 248$ (Bob)
- **compute shared session key as:**
- $K_{AB}=y^{x_A} \bmod 353 = 248^{97} \bmod 353 = 160$ (Alice)
- $K_{AB}=y^{x_B} \bmod 353 = 40^{233} \bmod 353 = 160$ (Bob)

Key Exchange Protocols

- Users could create random private/public Diffie-Hellman keys each time they communicate
- Users could create a known private/public Diffie-Hellman key and publish in a directory, then consulted and used to securely communicate with them
- Vulnerable to Man-in-the-Middle-Attack
- Authentication of the keys is needed

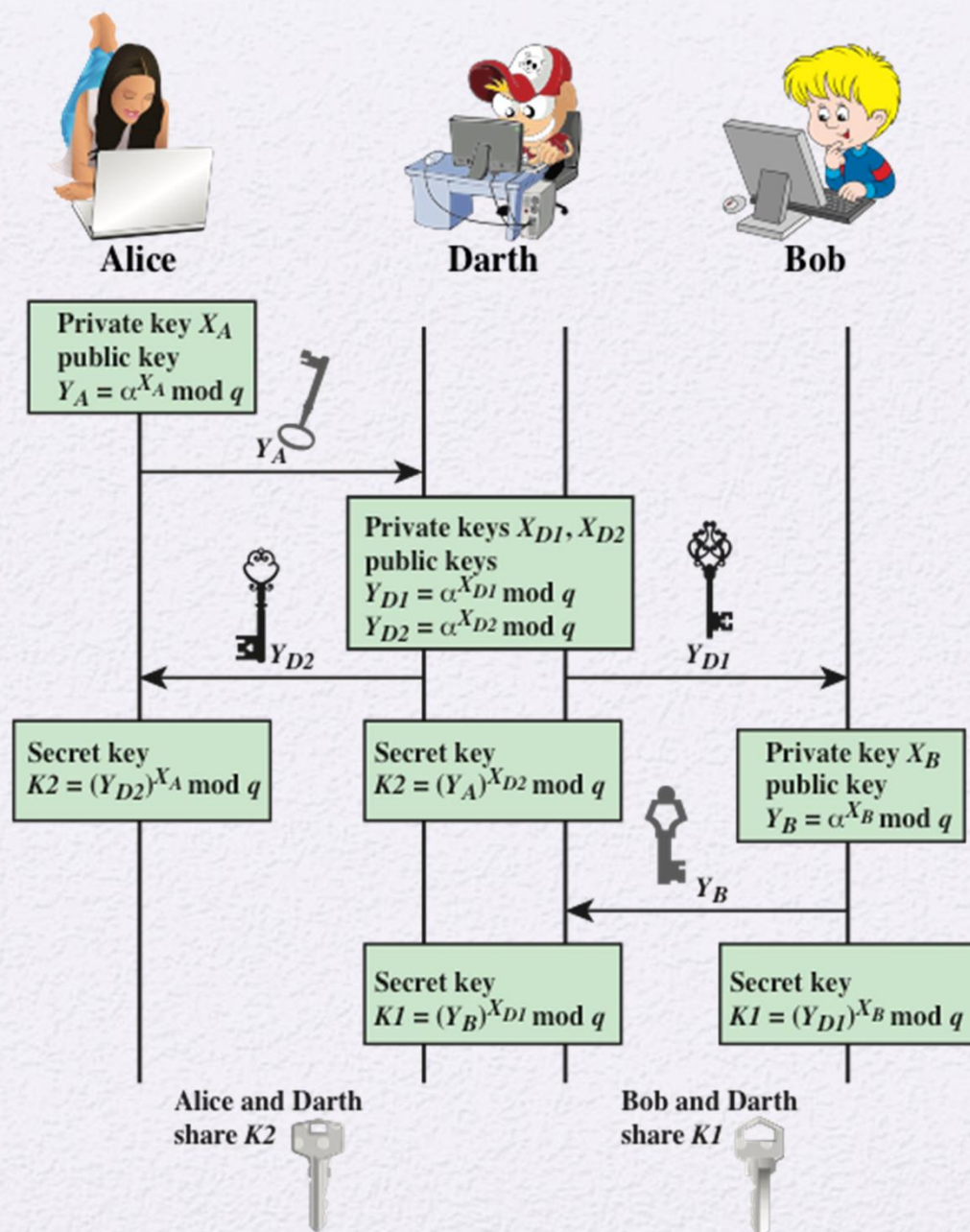


Figure 10.2 Man-in-the-Middle Attack